Journal of Quantitative Criminology
https://doi.org/10.1007/s10940-026-09671-9

ORIGINAL PAPER

Predicting behavioral health-involved police encounters: A machine learning approach

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Received: 17 June 2024 / Accepted: 18 March 2026
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Abstract

Objectives We evaluate the predictive accuracy of machine learning algorithms that forecast individual-level risk of behavioral health-involved encounters with police (BHIP) among community members with prior contact with first responders.

Methods We linked arrest and ambulance data from one city (May 2016-October 2017) for 199,679 unique individuals. The first 12 months provided predictive features; the subsequent 6 months served as the outcome window. We compared machine learning models utilizing 283 features to a baseline Prior High-User (PHU) approach based solely on prior BHIP counts.

Results Machine learning achieved positive predictive values (PPV) of 50.8%, 20.2%, and 7.5% when identifying the top 0.1% (n=199), 1% (n=1,996), and 5% (n=9,983) highest-risk individuals, representing improvements of 20%, 57%, and 44% over PHU. Critically, 28% of individuals in the ML-identified top 1% had no prior BHIP during the identification period, enabling true prevention. ML models demonstrated superior sensitivity, identifying 26.7% of all future BHIP events when targeting the top 1%, versus 17.1% for PHU. A simplified four-feature model using ambulance data maintained comparable performance. ML's improved PPV increases statistical power for intervention studies: a 200-person trial using ML-identified participants could detect a 38% reduction in BHIP, while achieving equivalent power with PHU would require 310 participants (55% larger sample).

Conclusions Machine learning substantially improves identification of individuals at BHIP risk, including those without prior events. This enhanced PPV can enable more efficient targeting of preventive services and make rigorous intervention research more feasible by increasing statistical power. Simplified models using only medical-services data suggest practical implementation pathways, though findings should be validated in other jurisdictions to ensure generalizability.

Keywords Machine learning · Forecasting · Behavioral health · Police encounters · Homelessness

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Published online: 15 April 2026

Introduction

Five percent of U.S. adults experience serious mental illness (SMI) annually. Yet a reported 25% of police shootings involve a person experiencing SMI or related behavioral health challenges (Fuller et al. 2015), with later research indicating even larger proportions involving persons with mental health disorders (Khan et al. 2024). Such disproportionate incidents indicate a potential need to better-manage behavioral health-involved encounters with police (BHIP). The Crisis Intervention Team model, for example, was developed to train officers to identify when individuals they encounter are experiencing behavioral health (BH) crisis, and to apply time-distance-cover de-escalation principles in addressing these events (Compton et al. 2014a, b, 2011, 2019; Watson and Compton 2019; Watson et al. 2017, 2021).

A key limitation of such approaches arises from their emphasis on managing current events rather than addressing prevention or long-term needs of individuals who experience these events. Studies show that individuals involved in BHIP often experience repeated encounters, suggesting that crisis intervention alone may be insufficient for long-term stabilization (Akins et al. 2016; Bravo et al. 2022). To bridge this gap, complementary strategies such as Houston's Chronic Consumer Stabilization Initiative (CCSI) seek to identify people who face predictable risks of BHIP, and to connect them with appropriate services that may prevent future BHIP (Council of State Governments Justice Center 2020).

A central component of any proactive strategy is a mechanism to identify those at highest risk for future BHIP events, who can then be prioritized for preventive services. In practice, preventive programs have generally used a 'high-user' strategy, assuming that individuals with prior high utilization are most likely to experience future incidents (Tentner et al. 2019; Ash et al. 2001; Iovan et al. 2020; Michelen et al. 2006). In the case of BHIPs, such an approach would use a high count of past BHIPs as the primary marker to identify and rank individuals at risk for future events of this type. These approaches, though valuable, do not explicitly consider other risk factors. Moreover, these approaches, by definition, do not identify people who have never experienced such an encounter but who face identifiable and elevated risks of a future BHIP. This limitation is substantial: in our data, approximately two-thirds of all individuals who experience a BHIP in the outcome period had no recorded BHIP in the preceding 12-month identification period.

This paper demonstrates that machine learning (ML) can substantially improve upon this baseline 'high user' approach, offering a more precise tool for identifying at-risk individuals—including those with no recent history of such events. Improved identification brings two critical benefits. First, as in other criminal justice and public health domains, it allows for focused resource allocation (Pan et al. 2017; Bhatt et al. 2024; Butts et al. 2015): providers can efficiently focus scarce, resource-intensive interventions on those who face the greatest identified risks. This targeting approach implicitly assumes that high-risk individuals will benefit from intervention—an assumption we examine further in the Discussion given the limited evidence base on intervention effectiveness in this domain.

A second, under-appreciated benefit of improved risk identification lies in its capacity to bolster the statistical power of program evaluations, thereby facilitating the development of a more robust evidence base. This is particularly important because the evaluation of programs designed to improve outcomes for people who experience BHIP has been hampered by a nascent evidence base (Compton et al. 2008; Dee and Pyne 2022; Watson et al. 2008). Rigorous evaluations are often hindered by the combination of high interven-

tion costs, which constrain sample sizes, and the relative rarity of BHIP as an outcome. Together, these factors lead to studies with low statistical power, making it difficult to detect true program effects. For instance, a recent randomized controlled trial of a police-mental health co-response team found the intervention was not significantly more effective than a traditional police response, with the authors noting that a smaller-than-intended sample size limited their ability to detect effects (Lowder et al. 2024). As we demonstrate in this paper, ML-based risk predictions help address this challenge by identifying individuals for trial recruitment who face the highest baseline risk of a future BHIP, thereby potentially increasing a study's baseline event rate and substantially increasing statistical power. Realizing these statistical power benefits depends on successfully recruiting and retaining identified high-risk individuals—which could pose implementation challenges given potential stigma and trust barriers.

We utilize comprehensive administrative data from [masked] containing detailed information on emergency responses over 18 months (May 2016-October 2017). We compare machine learning algorithms—specifically L1-penalized Logistic Regression, Random Forest, and LightGBM against a baseline Prior High-User (PHU) approach that ranks individuals solely by prior BHIP counts. The PHU-BHIP approach, despite its simplicity, proves remarkably effective in identifying persons at the most extreme risk, demonstrating that past behavior is indeed a strong predictor of future encounters, particularly for those with the most frequent recent history of BHIP. However, PHU-BHIP's effectiveness diminishes substantially when targeting larger populations beyond the highest utilizers.

To address these specific limitations while building on PHU's clear predictive signal, our ML models incorporate 283 engineered features from demographic, historical, and text-derived data to capture more nuanced patterns of risk. Beyond structured data, we systematically convert qualitative details from paramedic narratives into predictive flags capturing attributes like homelessness, behavioral health issues, and substance use. This approach allows us to retain PHU-BHIP's ability to identify high-risk individuals with established patterns while extending prediction capability to those without prior events.

The use of machine learning in justice settings rightly raises concerns about fairness and potential algorithmic bias, particularly when models include sensitive demographic and police arrest data (Dressel and Farid 2018; Rudin et al. 2020; Council on Criminal Justice 2024). We address these concerns through two sets of analyses. First, our analysis of feature importance shows that arrest history and demographic features have negligible impact on model predictions. Instead, predictive power is driven almost entirely by text-derived flags from paramedic narratives and the timing of prior ambulance events. Second, we conduct a detailed subgroup analysis across racial categories and find that our machine learning models achieve relative parity on common operationalizations of key fairness metrics (Chouldechova 2017). These fairness analyses reveal that a large number of features contribute negligibly to predictions, suggesting that simpler, more interpretable models may be feasible without sacrificing accuracy.

This finding addresses another critical concern in justice settings: the 'black box' nature of complex ML models, which creates barriers both for practitioners seeking to understand and validate predictions and for individuals who receive risk scores but cannot understand the basis for their assessment. Algorithmic transparency is increasingly recognized as essential for building trust, enabling oversight, and ensuring accountability in high-stakes decisions (Rudin 2019). Our analysis directly addresses this challenge by demonstrating that

3- and 4-variable models can maintain most of the accuracy gains of the full ML approach while providing the transparency needed for justice settings.

This study makes several contributions to the literature on behavioral health crisis prediction and prevention. First, we demonstrate that machine learning approaches substantially improve upon traditional high-user identification methods in our setting, achieving 20-57% improvements in positive predictive value across different risk thresholds. Second, we show that ML models can identify individuals at risk of a BHIP who have no recent history of such events. In our data, fully 28% of individuals identified by the ML in the top 1% risk group had no recorded BHIP during the 12-month identification period, representing substantial preventive potential. Third, we develop a systematic methodology for creating interpretable risk assessment tools by combining SHapley Additive exPlanations (SHAP)-based feature importance analysis with FasterRisk optimization, producing memorizable 3- and 4-feature scoring tools that maintain much of the predictive performance of complex models while ensuring transparency suitable for justice settings. Fourth, we demonstrate how enhanced prediction accuracy can transform intervention research by reducing required sample sizes by 33-55% for detecting equivalent treatment effects.

Finally, our study demonstrates that effective prediction for prevention depends less on algorithmic sophistication than on measurement quality. Our results strongly suggest that performance gains stem primarily from converting rich clinical observations into structured measures. This finding has important implications for implementation: jurisdictions need not invest heavily in cutting-edge prediction technology but may benefit from developing robust data integration and text processing capabilities that leverage the expertise of front-line staff. The consistency in performance across different algorithmic approaches—coupled with the strong results from simplified scoring rules—suggests that the primary challenge lies in systematically capturing and structuring the rich clinical observations already being documented by paramedics and other first responders.

The paper proceeds as follows: Section 2 reviews relevant literature on police-behavioral health encounters and predictive modeling approaches; Section 3 details our data sources, record linkage methodology, feature engineering, and model evaluation framework; Section 4 presents our main findings on predictive performance, feature importance, fairness metrics, and statistical power implications; and Section 5 discusses implications for policy and practice while acknowledging limitations and directions for future research.

Background

The risks attendant to interactions between law enforcement and individuals experiencing behavioral health crises has garnered increased attention in recent years. Some researchers argued that deinstitutionalization policies produced a significant increase in the number of visible encounters between law enforcement and persons living with serious mental illness (SMI) (Lamb & Weinberger, 1998; Teplin, 1984). Others suggest that this shift has resulted in the criminal justice system becoming a *de facto* mental health provider, a phenomenon known as the "criminalization of mental illness" (Abramson 1972; Engel and Silver 2001; Lamb and Weinberger 2005; Pope et al. 2024). Persons living with SMI are overrepresented in the criminal justice system (Skeem et al. 2011; Steadman et al. 2009; Teplin 1990). This overrepresentation has been attributed to various factors, including lack of adequate

community-based mental health services, the increased use of police to manage individuals in crisis, and the difficulty of navigating the complex criminal justice system (Fisher et al. 2006; Lamb and Weinberger 2013). Although people who experience behavioral health challenges may share some risk factors for recidivism (e.g., history of contact with the criminal justice system and challenging social relationships) with other persons who interact with the criminal justice system, people who live with behavioral health problems also face unique factors likely to increase rearrest (e.g., increased surveillance when symptomatic and social stigma), and they disproportionately experience homelessness (Balyakina et al. 2014; Herbert et al. 2015; Jacobs and Gottlieb 2020; Jacobs et al. 2024; Morabito and Socia 2015; Moschion and Johnson 2019).

The nature of police interactions with individuals experiencing behavioral health crises has been a focus of criminological research. A growing literature examines the use of force by police during these encounters (Engel et al. 2020; Morabito et al. 2012; Morabito and Socia 2015; Rossler and Terrill 2017; Pope et al. 2024) as well as the role of specialized police responses, such as the Crisis Intervention Team (CIT) model (Taheri 2016; Wood et al. 2017). The actual effectiveness of CIT and other interventions in reducing the overall prevalence of behavioral health-involved police encounters (BHIPs) remains unclear (Patterson and Swan 2019; Peterson and Densley 2018; Watson and Wood 2017). More recent co-responder models—pairing officers with mental-health clinicians—have shown greater success at diverting people from emergency departments, reducing arrests and involuntary transports (Petreca et al. 2025; Davis et al. 2025), while qualitative work highlights the importance of shared data systems and joint training for these teams to function effectively (Fisher et al. 2024). Denver's STAR program dispatched non-police clinicians to low-risk mental health calls, yielding a 34% drop in low-level crime reports without increasing serious crime, at a fraction of typical police-response cost (Dee and Pyne 2022). Similar strong benefits were identified for Eugene, Oregon's Crisis Assistance Helping Out on the Streets (CAHOOTS) mobile crisis team model (Davis et al. 2025). Still, there is a growing recognition of the need for more proactive, preventive approaches to address the underlying risk factors for BHIPs.

One approach to identifying individuals at risk of future BHIPs is to focus on "high users" of emergency services (Tentner et al. 2019). This approach aligns with the broader literature on targeting interventions towards individuals with a history of frequent service utilization (Boland et al. 2023; Burton et al. 2022; Iovan et al. 2020). However, traditional high utilizer approaches may not fully capture the complex risk factors associated with BHIPs and may miss opportunities for early intervention and prevention.

Another promising avenue for identifying individuals at risk of future BHIPs is through the application of machine learning techniques (Ridgeway 2013) that leverage rich individual-level data sources. ML has been used in criminological research to predict various outcomes, such as recidivism (Barnes et al. 2017; Berk 2017; Laqueur and Copus 2024), domestic violence (Arora et al. 2024; Berk et al. 2016) and police misconduct (Helsby et al. 2018). These studies have demonstrated the potential of ML to improve risk prediction in criminal justice settings. The use of ML to predict BHIPs remains an emerging area of research. The use of ML in high-stakes decision-making also raises important ethical and legal concerns–including concerns about algorithmic bias and fairness (Berk et al. 2021; Lum and Isaac 2016).

This study addresses several gaps in the existing literature. While previous research has focused primarily on improving police response during behavioral health encounters, there has been limited work on developing systematic, data-driven approaches for identifying individuals at risk of future BHIPs before they occur. Moreover, existing high-user approaches, while valuable, may miss opportunities for early intervention among individuals without prior encounter history. To address these limitations, we develop and evaluate machine learning models using comprehensive administrative data from [masked] to predict future BHIP risk, comparing their performance against traditional high-user approaches while systematically examining issues of algorithmic fairness and practical implementation.

Empirical Methodology

Data

We obtained Fire Department (FD) administrative data on all 379,179 ambulance runs with patients on-scene between May 2016 and October 2017. Each record contained information on event date, time, and location, and individual patient data including name, date of birth, gender, race, and address. Records also include rich semi-structured and unstructured text fields entered by FD medics (e.g., patient complaint, patient symptoms, medic impression, event type, and event narrative). Employing keyword searches, we use these fields to classify events into 56 diverse categories listed in Table 1. These categories capture incident attributes which may be predictive of future BHIP such as indications of homelessness, substance use, or behavioral crises. We also obtained PD arrest data for all 100,592 arrests over the same period. Each record details the arrest date, time, and location, and arrestee demographic characteristics. Arrest records also indicate charging offense and associated FBI code (e.g. 4A: simple assault).

We use all 18 months of ambulance and arrest data, split into a 12-month identification period (May 2016-April 2017), and a six-month (May-October 2017) outcome period. We applied a probabilistic record linkage algorithm (Tahamont et al. 2021) to link records belonging to the same individual within and across the ambulance and arrests datasets (See Appendix).

This process resulted in a final analytic sample of 199,679 unique individuals who experienced at least one ambulance or arrest event during the identification period of May 2016-April 2017. Of these, 160,485 (80.4%) experienced at least one FD ambulance event, and 50,242 (25.2%) experienced at least one arrest in this period. It is important to note that during this study period, the jurisdiction did not have a formal, city-wide 'high-user' identification program for this population; the 'status quo' relied on traditional emergency response rather than the kind of proactive targeting explored in this paper.

For each person, we capture demographics and compute detailed metrics on contacts during the identification period. Roughly 3% of the full sample (n=5,583) experienced a BHIP during the identification period. Our main outcome is whether individuals experienced at least one BHIP in the 6 month outcome period. More precisely, our outcome captures whether an ambulance run in the outcome period involved an individual with indications of behavioral health concerns and indicated police presence on scene. Among the individuals in the identification period 0.8% (n=1,512) went on to have a BHIP in the May-October

Table 1 Text-derived feature flags and associated search terms

Flag	Search Term	Prev. (%)	Flag	Search Term	Prev. (%)
BH†	agitated, anxiety, behavioral, combative, crying, depression, distressed, fear, hallucination, hostile, irritable, nervous, panic, psychiatric, stress, suicidal, taser injury, tearful, violent	10.2	Denies	denies	46.7
Police†	[masked]pd, police, [masked].p.d, [masked] p d	11.2	No evidence	no evidence	1.0
BH-police†	The union of keywords for the BH and police flags	2.8	Heart	chest, heart, cardiac	28.4
Homeless	homeless, homless, undomicile, without home, without residence, no home, no residence, nfa	1.9	Sleep	sleep, insomnia	3.8
Suicide	suicide, suicidal	0.5	Meds	meds medication, meds	8.3
Depression	depression	1.3	Substance	substance abuse, overdose	10.4
Anxiety	anxiety	6.8	Homicidal	homicidal	0.5
Agitated	agitated	5.1	Shake	shake, shaking	1.7
Violent	violent	0.6	Nausea	nausea, vomit	17.5
Voices	voices, hallucination, delusion	0.9	Paranoid	paranoid	0.2
Aggressive	aggressive, hostile, combative	2.7	Psych	psych	5.2
Alcohol	alcohol, etoh, intox, odor, slur	23.6	Taser	tazer, taser	0.1
Cold	cold cold, chills, shiver, numb	6.1	Slow	slow	3.0
Pain	pain, hurt, injur	58.0	Sweat	sweat	1.2
Nervous	nervous	1.0	Swell	swell	5.1
Confused	confused confuse, disorient	5.9	Request ER	request er, requesting er	6.4
Breathing	breath, breathing, hyperventilate	29.2	Altered mental	altered mental	4.6
Feel right	feel right do not feel right	2.8	Cut	laceration, cut	8.7
Upset	upset, irritable	26.7	Refuse	refuse, refusing	14.5
Restless	restless	0.5	Rule out	rule out	8.2
Hungry	hunger, hungry	0.2	Seizure	seizure	5.3
Stress	stress	25.6	Trauma	trauma	24.8
Bizarre	bizarre	1.1	Weak	weak	10.1
Bleed	bleed	10.3	Diabetic	diabetic, glycemia, diabetes	4.0
Difficulty	difficulty, cant, can't	21.9	Faint	faint, unconscious	9.2
Cry	cry, tear	2.3			

Note: Table shows flags derived from text fields in ambulance records. First column of each panel shows the category, second column lists comma-separated words and phrases used for identification, and third column shows the prevalence of each flag. Flags marked with a dagger (†) are used in defining the outcome variable

2017 outcome period. Appendix Fig. 3 provides a flow diagram illustrating the sample construction from initial data sources to the final linked dataset. The first column of Table 5 shows the baseline demographic characteristics and service encounter history for the full analytic sample. All data use and storage are approved by [masked IRB].

Measuring the Outcome

To define and measure our outcome, we rely on text fields compiled by ambulance staff for each run, which capture information on patient complaint, symptoms, and medic impressions. We worked closely with FD staff in developing logic to flag BH-involved events, and those with indicated police presence (See Appendix D). A primary finding that motivates our outcome measurement strategy is that police arrests are an insufficient proxy for police presence in BH-related events. Across all ambulance runs, we identified 40,593 (11%) with an indication of police presence in the narrative data. However, a much smaller 2,169 of these runs—5.3% overall—were associated with an arrest. Relying solely on arrest data would therefore cause an analysis to miss the vast majority of police encounters and understate the true police footprint in behavioral-health-involved calls.

An analysis of incident characteristics indicates that ambulance calls with police present differ significantly from those without. Appendix Fig. 4 shows that encounters with police are far more likely to involve indications of behavioral crises, substance use, and aggression. For example, events with police presence are more than twice as likely to have indications of a behavioral health issue (26.1% vs. 9.7%), alcohol use (39.9% vs. 23.5%), or general substance use (47.6% vs. 28.8%). Indicators of agitation (17.9% vs. 3.5%) and aggression (12.7% vs. 2.1%) are also substantially more frequent when police are on scene. Conversely, Appendix Fig. 4 also shows that calls without police presence are more likely to involve traditional medical complaints such as pain (66.5% vs. 56.0%), cardiac issues (30.8% vs. 20.2%), and breathing difficulties (32.3% vs. 23.3%). This affirms that police presence is a strong marker for a distinct type of event where behavioral and safety concerns are salient.

Predictive Features

While the PHU method relies on a single predictive feature — the count of an individual's past BHIPs during the identification period — our ML models incorporate a comprehensive set of 283 individual-level features derived from linked ambulance and arrest data. Reflecting an exploratory, data-driven strategy, our approach to feature engineering favored breadth, aiming to provide the ML algorithm with a rich and diverse set of potential predictors rather than pre-selecting features based on strong theoretical assumptions. The intention was to allow the model itself to discern the most salient indicators of BHIP risk from this extensive feature set.

To assist in interpretability of our feature generation approach, we group these 283 features into six mutually exclusive categories:

- **Demographics**: To ensure consistency for individuals with multiple contacts, we use the modal (most frequently recorded) value for gender and race. We create these demographic features using information from both ambulance and arrest data.
- **PD-Derived**: Features derived exclusively from Police Department (PD) arrest data, including detailed arrest counts by charge class (e.g. misdemeanor, felony, etc.) and 28 FBI crime codes, the timing of first and last arrests, and arrest persistence over the identification period, measured as the number of unique months in which an individual had at least one arrest.

- **FD-Derived (Non-Text)**: Features from structured fields in the Fire Department (FD) ambulance data, such as the total count of ambulance runs, timing of first and last runs, and persistence.
- **FD-Derived (Text)**: A significant portion of our features are derived from keyword searches on text fields in FD ambulance records, resulting in 56 distinct flags. For each of these flags, we systematically generate features that mirrors our approach to service contact history, capturing the total count, the time since most recent flag, and the time since the flag's first appearance. This category also includes composite summary indicators for mental health, substance use, and aggression, as well as three specific flags used to define our study's outcome: BH-involved, police-involved, and BHIP (their co-occurrence).
- **Combined Persistence**: A variable measuring the chronicity of service contact across both agencies by counting the number of unique months a person had any contact.
- **Modal Temporal Patterns**: Features capturing the most common season and time of day for an individual's past encounters with both police and ambulance services.

Appendix Table 10 provides the counts for the feature groups described above.

Model Training

Our primary goal is to compare the efficacy of a machine learning approach against two simple and transparent Prior High-User (PHU) baselines (Howson et al. 2022; Roysden and Wright 2015; Yan et al. 2019; Yang et al. 2019). These heuristics represent a trade-off between predictive specificity and breadth.

The first baseline, PHU-BHIP, ranks individuals solely by their count of prior BHIPs during the identification period. This approach is highly specific to the outcome of interest but is definitionally unable to identify at-risk individuals who have not yet had such an encounter, limiting its use for true prevention.

The second baseline, PHU-AMB, ranks individuals by their total count of ambulance runs. This model is also simpler to implement from an administrative standpoint, as it relies on a direct count rather than the complex text analysis required to identify a BHIP. It captures a broader population of general service utilizers but is less specific to the BHIP outcome and may identify individuals with high medical needs rather than those at risk of a behavioral health crisis.

We compare these heuristics to three distinct and widely-used machine learning models: L1-penalized Logistic Regression, Random Forest (RF) (Breiman 2001), and LightGBM (Shi et al. 2022), a gradient-boosting framework. These models were selected to represent a range of standard techniques often applied to this type of prediction problem.[1]

We seek to generate a single set of out-of-sample risk scores for every individual in our study population for use as analytical variables in subsequent analyses. To accomplish this, we employ a nested cross-validation structure with $K = 2$ outer folds. We partition the data into two equally-sized, mutually exclusive folds, meaning that no individual is present in both folds. We first train a model on Fold 1 and use it to generate predictions for the indi-

[1] All analyses were conducted using Python. We implemented the L1-penalized Logistic Regression and Random Forest models using the *scikit-learn* library and the gradient boosting model using the *lightgbm* library

viduals in Fold 2. We then train an entirely new model on Fold 2 to generate predictions for individuals in Fold 1.

Within each of the two outer training folds, we perform hyperparameter tuning using 3-fold inner cross-validation. This inner cross-validation process systematically searches a pre-defined grid of parameters to find the combination that maximizes average performance on the heldout inner folds.[2] A final model is trained for each outer fold using the optimal hyperparameters and then out-of-sample predictions are generated for individuals in the other outer fold.

Unlike traditional machine learning applications that would discard these predictions and train a final model for deployment, we retain these out-of-sample predictions as research data. Having generated risk scores for all 199,679 individuals who had contact with our city's emergency response system during the study period, we can directly examine the characteristics of those who would be selected for intervention. This allows us to identify which individuals would be targeted by the algorithm, examine demographic patterns in risk assignment, and evaluate intervention strategies—all while maintaining proper separation between training and evaluation. When computing performance metrics such as positive predictive value at specific thresholds, we pool these out-of-sample predictions across both outer folds to obtain a single ranked list of all individuals.[3]

Beyond assessing overall predictive accuracy, we conduct two further sets of analyses to interpret model behavior and explore practical implementation trade-offs. First, to understand which features drive the models' predictions, we employ SHAP (SHapley Additive exPlanations) analysis (Lundberg and Lee 2017), a method that quantifies the contribution of each feature to the final risk score. This provides transparency into the mechanisms underlying the model's performance. Second, to address potential ethical concerns and implementation constraints, we perform a series of feature exclusion experiments. These experiments systematically test how model performance changes when sensitive data, such as arrest history and demographic features, are removed. This includes the evaluation of a highly simplified models (Liu et al. 2022) using only three or four features derived from ambulance data, allowing us to assess the trade-offs between predictive power, simplicity, and reliance on different data sources.

Results

We begin by directly comparing the predictive performance of our ML models against two Prior High-User baselines. We then deconstruct this performance, first by highlighting the ability of the ML approach to identify at-risk individuals without prior BHIP events, and second, by using feature importance analyses to reveal the specific data elements that drive predictions. We then turn to an analysis of who gets identified, presenting demographic

[2] For the L1-penalized Logistic Regression, we tuned the inverse of regularization strength (C: 0.01, 1, 1000). For the Random Forest model, we tuned the number of trees (n_estimators: 50, 100, 1000), maximum tree depth (max_depth: 2, 5, 10), and the proportion of features considered at each split (max_features: 0.1, 0.5). For the LightGBM model, we tuned the learning rate (0.01, 0.05, 0.1), the number of estimators (50, 100, 200), and the maximum depth (2, 5, 10).

[3] In Appendix F we show that pooling predictions before computing performance metrics generally yields lower variance estimates compared to averaging metrics across folds, while providing a more operationally relevant estimate.

characteristics of the individuals predicted to be at highest risk and comparing accuracy and errors rates across race groups. We conclude by demonstrating a key practical application of the model's enhanced PPV: its ability to substantially increase the statistical power of future program evaluations.

Prior-High-User versus ML approach

To evaluate the performance of the machine learning and baseline approaches, we compare their ability to identify individuals who will experience a future BHIP. Recognizing that practitioners may wish to target different-sized groups depending on intervention intensity (e.g. smaller populations for 'high-touch' interventions; larger populations for less-costly, 'light-touch' interventions) (Pan et al. 2017), we assess performance across several risk thresholds: the top 0.1% (N=199), 1% (N=1,996), 2.5% (N=4,992), and 5% (N=9,983) of individuals as ranked by each model. We choose these thresholds to represent a tiered approach to intervention, where service intensity is matched to the level of predicted risk. For example, the most selective group (the top 0.1%) could be targeted for high-touch, resource-intensive interventions like Assertive Community Treatment, intensive case management, or supportive housing programs. A larger group (the top 1%) might receive services such as enrollment in a specialized care coordination program, frequent follow-ups from community paramedics, or facilitated access to substance use disorder treatment. The broadest group we examine (the top 5%) could be suitable for light-touch preventive strategies implemented at scale, such as proactive outreach from community health workers, provision of information about support services, or enhanced connections to integrated primary care.[4]

We use two key metrics for this evaluation: *positive predictive palue* (PPV), which captures the proportion of individuals flagged as high-risk who actually have a future BHIP, and *sensitivity*, which measures the model's reach (i.e., the proportion of all individuals with future BHIPs that were identified as high-risk). When necessary, we break risk-score ties at the cut-off through random selection (See Appendix Section E for minimum and average identification-period counts for PHU-identified individuals.).

Table 2 presents the results of this comparison, evaluating the three machine learning models against the two PHU baselines. The results highlight the trade-offs between the simple heuristics and the more complex ML approach. The highly specific PHU-BHIP model yields a PPV of 0.422 for the most selective 0.1% risk group, demonstrating that a simple count of prior events can identify a population with highly-elevated risk. However, its PPV decreases sharply to 0.129 for the top 1% and 0.052 for the top 5%. The broader PHU-AMB baseline, which is less specific to the outcome, shows consistently lower PPV and sensitivity across all thresholds.

In contrast, the machine learning algorithms generally demonstrate superior performance. The LightGBM model, for instance, achieves a PPV of 0.508 for the top 0.1% of individuals, 0.202 for the top 1%, and 0.075 for the top 5%. Similarly, the Random Forest

[4] These thresholds are also chosen in consideration of the observed distribution of prior BHIPs in our data. Given that approximately 2.8% of individuals in our sample experienced one or more BHIPs during the identification period, extending intervention targets beyond this natural prevalence with a PHU approach (such as to the top 5% risk group explored in our results) inherently involves selecting individuals with no prior BHIP history.

Table 2 Grouped performance of machine learning models and prior high-user baselines

Model	Top 0.1%	Top 1%	Top 2.5%	Top 5%
PPV				
LightGBM	0.508	0.202	0.117	0.075
	(0.437-0.583)	(0.183-0.220)	(0.108-0.126)	(0.069-0.080)
Random Forest	0.492	0.196	0.117	0.075
	(0.422-0.558)	(0.179-0.214)	(0.108-0.127)	(0.070-0.080)
Logistic Regression	0.392	0.190	0.114	0.076
	(0.337-0.472)	(0.173-0.207)	(0.105-0.123)	(0.070-0.081)
Prior High-User (BHIP)	0.422	0.129	0.092	0.052
	(0.347-0.503)	(0.116-0.144)	(0.086-0.101)	(0.048-0.057)
Prior High-User (AMB)	0.261	0.109	0.073	0.048
	(0.201-0.322)	(0.096-0.123)	(0.066-0.080)	(0.044-0.052)
Sensitivity				
LightGBM	0.067	0.267	0.387	0.493
	(0.058-0.076)	(0.246-0.287)	(0.363-0.410)	(0.469-0.517)
Random Forest	0.065	0.259	0.388	0.493
	(0.056-0.073)	(0.238-0.279)	(0.364-0.411)	(0.469-0.518)
Logistic Regression	0.052	0.251	0.376	0.499
	(0.044-0.062)	(0.231-0.271)	(0.354-0.401)	(0.475-0.525)
Prior High-User (BHIP)	0.056	0.171	0.305	0.345
	(0.046-0.066)	(0.154-0.191)	(0.285-0.329)	(0.324-0.369)
Prior High-User (AMB)	0.034	0.144	0.241	0.317
	(0.026-0.042)	(0.128-0.161)	(0.220-0.262)	(0.296-0.340)

Note: Table displays the comparative performance of the LightGBM, Random Forest, Logistic Regression, and Prior High-User baselines (BHIP and AMB) models in predicting Behavioral Health-Involved Police (BHIP) encounters. For each model, both Positive Predictive Value (PPV) and Sensitivity are reported at various identification thresholds: the top 0.1%, 1%, 2.5%, and 5% of individuals identified as highest risk. The primary entries for each model and threshold represent the point estimates, with the corresponding 95% bootstrapped confidence intervals (1,000 replications) presented in parentheses below

model shows comparable results with PPVs of 0.492, 0.196, and 0.075 for the respective thresholds. The Logistic Regression model also generally outperforms PHU, yielding PPVs of 0.392, 0.190, and 0.076. These figures represent statistically significant improvements over the PHU-BHIP baseline across most thresholds, highlighting the enhanced efficiency of ML approaches in identifying individuals at highest BHIP risk, especially when identifying individuals outside of the top 0.1%.

The improvements in PPV also lead to improvements in sensitivity for the ML models. For the top 1% of individuals, the LightGBM model identifies 26.7% of all individuals who will go on to experience future BHIPs, the Random Forest model identifies 25.9%, and the Logistic Regression model identifies 25.1%. This is substantially higher than the 17.1% achieved by the PHU-BHIP approach at the same threshold. At the 5% threshold, the ML models (LightGBM: 49.3%, Random Forest: 49.3%, Logistic Regression: 49.9%) identify a much larger proportion of individuals who will go on to experience future BHIP events compared to the PHU-BHIP model (34.5%). These sensitivity results underscore the greater reach and efficiency of ML in identifying individuals who will experience a future BHIP.

We find remarkable consistency in performance across the LightGBM, Random Forest, and Logistic Regression models. This similarity suggests that the documented improve-

ments are not an artifact of a particular algorithm, but are instead driven by the predictive information contained in our feature set. As we will demonstrate in the subsequent feature importance and exclusion analyses, this predictive power stems overwhelmingly from the features engineered from the text of ambulance narratives, rather than from demographic or police arrest data. Given this consistency, and for the sake of brevity and clarity, we will primarily report results from the LightGBM model in subsequent analyses, as its performance is representative of the ML approach as a whole.[5]

Performance for Individuals with and without Recent BHIP History

A critical aspect of a proactive risk identification strategy is the ability to flag individuals who may not have had a behavioral health-involved encounter with police in the recent past, yet face identifiable elevated risk for a future event. This is particularly important given that a substantial percentage—approximately 67.2% of all individuals in the prediction sample who experience a BHIP in the outcome period—had no recorded BHIP in the 12-month identification period.

Table 3 shows PPV and Sensitivity for the LightGBM and PHU-BHIP models across four risk thresholds.[6] The table is divided into two sections to show performance for individuals with at least one BHIP during the identification period and those with no prior BHIPs. For the subgroup of individuals with a prior BHIP, the LightGBM model yields a PPV of 0.51 at the top 0.1% threshold and 0.12 at the 5% threshold. For the same subgroup, the PHU-BHIP model's PPV is 0.42 at the 0.1% threshold and 0.09 at the 5% threshold. In terms of sensitivity for this group, the LightGBM model's values range from 0.20 at the top 0.1% to 0.95 at the 5% threshold; the PHU-BHIP model's sensitivity ranges from 0.17 to 1.00 across the same thresholds.

The performance within the subgroup of individuals who have no prior BHIPs highlights an important advantage of the machine learning approach. The LightGBM model demonstrates a capacity for true prevention by identifying individuals at risk of a first BHIP (within the last year)—a task for which the PHU-BHIP model is definitionally unsuited. For example, the ML model identifies individuals in this "no priors" group with a PPV of 0.1 at the 1% threshold.

It is important to contextualize these results. While the machine learning model represents an improvement in identifying at-risk individuals without a prior BHIP, the lower PPV for this subgroup compared to those with a known history highlights an important reality: this group is inherently more difficult to predict. This is clearly illustrated by the difference in model sensitivity. For example, at the 5% risk threshold for individuals with a prior BHIP, the LightGBM model recalls 95% of individuals with future BHIPs. In contrast, for the 'no priors' group, the model's sensitivity is far lower at 27%. While this represents a substantial improvement over a baseline that finds almost none, it also underscores the challenge of true prevention, as most persons who will experience their first BHIP are not captured even in the ML-identified high-risk group.

[5] Full results from all models are available in the Appendix.

[6] Appendix Table 11 shows the counts associated with each model and threshold.

Table 3 Model performance for subgroups with and without prior BHIPs

Model	Top 0.1%	Top 1%	Top 2.5%	Top 5%
With Prior BHIP				
PPV				
LightGBM	0.51	0.22	0.14	0.12
	(0.43-0.58)	(0.20-0.25)	(0.13-0.16)	(0.10-0.13)
FasterRisk	0.50	0.21	0.11	0.09
	(0.42-0.57)	(0.18-0.22)	(0.10-0.11)	(0.08-0.10)
PHU-BHIP	0.42	0.13	0.09	0.09
	(0.35-0.50)	(0.12-0.14)	(0.08-0.10)	(0.08-0.10)
Sensitivity				
LightGBM	0.20	0.69	0.88	0.95
	(0.17-0.23)	(0.65-0.73)	(0.85-0.91)	(0.93-0.97)
FasterRisk	0.19	0.71	0.95	1.00
	(0.16-0.22)	(0.67-0.74)	(0.93-0.97)	(1.00-1.00)
PHU-BHIP	0.17	0.52	0.93	1.00
	(0.14-0.20)	(0.48-0.57)	(0.91-0.96)	(1.00-1.00)
	(199-199)	(1,996-1,996)	(4,991-4,991)	(5,433-5,733)
Without Prior BHIP				
PPV				
LightGBM	0.50	0.13	0.08	0.05
	(0.12-1.00)	(0.10-0.16)	(0.06-0.09)	(0.04-0.05)
FasterRisk	0.17	0.15	0.12	0.04
	(0.00-0.60)	(0.11-0.19)	(0.09-0.15)	(0.04-0.05)
PHU-BHIP	N/A	N/A	N/A	0.01
				(0.00-0.01)
Sensitivity				
LightGBM	0.00	0.06	0.15	0.27
	(0.00-0.01)	(0.05-0.08)	(0.13-0.17)	(0.24-0.30)
FasterRisk	0.00	0.04	0.06	0.19
	(0.00-0.00)	(0.03-0.05)	(0.05-0.08)	(0.17-0.22)
PHU-BHIP	N/A	N/A	N/A	0.02
				(0.02-0.03)

Note: For each subgroup, the table shows the Positive Predictive Value (PPV) and Sensitivity from the top-ranked group across various identification thresholds (Top 0.1%, 1%, 2.5%, and 5%). Point estimates are presented, with 95% confidence intervals from 1,000 bootstrap replications shown in parentheses. The PHU model by definition does not identify individuals without prior BHIPs at certain thresholds

Feature Importance

To understand the mechanisms underlying our model's predictions, we conducted a SHAP (SHapley Additive exPlanations) analysis on the LightGBM model. SHAP is a game-theoretic approach that assigns each feature an importance value for every individual prediction by calculating the marginal contribution of that feature across all possible feature combinations (Lundberg and Lee 2017). Unlike traditional feature importance methods that provide only global rankings, SHAP values offer both local (individual- or subgroup-level) and global (population-level) interpretability. For our analysis, SHAP values represent the change in log-odds of BHIP risk attributable to each feature. Positive SHAP values indicate features that push the model toward predicting higher risk, while negative values indicate

protective factors. By examining mean SHAP values across our population, as well as subsets of features and individuals, we can identify which features most strongly influence the model's risk assessments for which individuals.

Figure 1 presents these mean SHAP values with features organized hierarchically to reveal patterns in predictive importance. Features are first grouped by data source (ambulance vs. arrest), then by feature type (counts/history vs. temporal/recency patterns), and further distinguished by whether they derive from structured data fields or text flags. Within text-derived features, we separately display specific behavioral flags (e.g., mental health, substance use, aggression) to understand which clinical indicators drive predictions. This hierarchical organization allows us to assess not just which individual features matter, but whether predictive power comes from criminal justice involvement versus medical encounters, from raw service utilization versus temporal patterns, and from structured administrative data versus clinical narratives. The analysis is further stratified by both predicted risk level (top 5% vs. all others) and prior BHIP status to examine whether the model relies on different features for different subpopulations.

The analysis reveals three findings:

First, text-derived flags from ambulance narratives dominate the model's decision-making process. The six most influential features—ambulance history, behavioral health flags, psychiatric indicators, mental health flags, police involvement flags, and substance use indi-

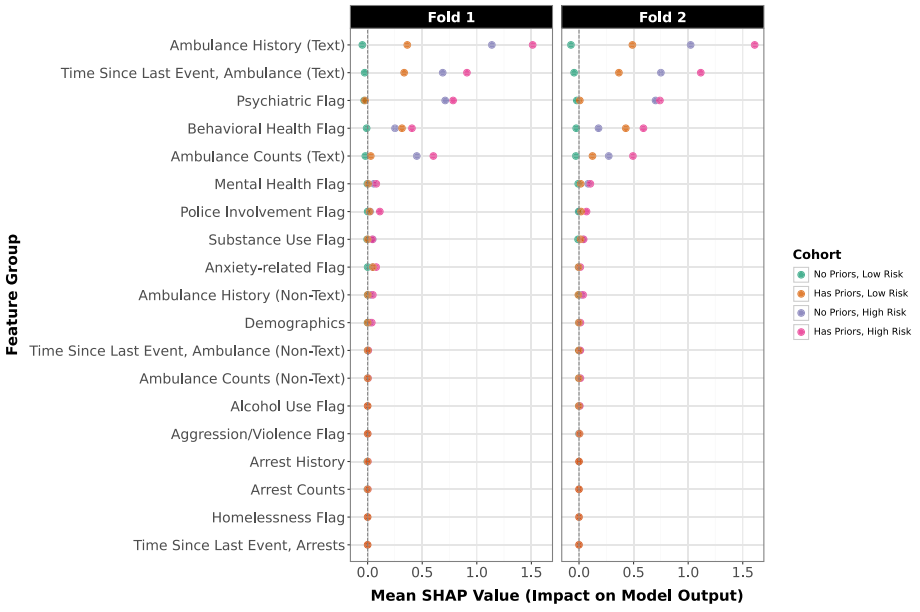


Fig. 1 Mean SHAP (SHapley Additive exPlanations) values for LightGBM model features, stratified by risk and prior BHIP status. This plot illustrates the average impact (in log odds) of different feature groups on the LightGBM model's output. Features are grouped by their type and ranked by their overall mean SHAP value. The plot is split into two folds (Fold 1 and Fold 2) to show consistency of feature importance across different data partitions. Within each fold, points are colored to represent four distinct cohorts: 'No Priors, Low Risk' (green), 'Has Priors, Low Risk' (teal), 'No Priors, High Risk' (purple), and 'Has Priors, High Risk' (orange). 'Priors' refers to whether an individual had a prior BHIP encounter during the identification period

cators—are all derived from paramedic text fields. In contrast, arrest-related features (arrest counts, arrest history, time since arrest) and demographic variables cluster at the bottom of the importance ranking with near-zero mean SHAP values. This pattern holds consistently across both folds, demonstrating that our model's predictive power stems from behavioral indicators documented by medical professionals rather than criminal justice involvement or demographic characteristics.

Second, the SHAP analysis illuminates how the model identifies at-risk individuals without prior BHIP history. For individuals without prior BHIP (green and purple markers in Fig. 1), text-derived behavioral health and substance use flags still generate substantial SHAP values, though lower than for those with prior history. This indicates the model detects emerging risk signals in ambulance narratives—such as mental health concerns, substance use, or crisis indicators—before an individual's first police-involved behavioral health encounter. The temporal features (time since last ambulance event) also contribute meaningfully for this group, suggesting that recent ambulance contact patterns help identify escalating risk. This mechanistic insight explains our finding that 28% of individuals in the model-identified top 1% risk group had no prior BHIP during the identification period.

Finally, the minimal contribution of arrest and demographic data to the model's predictions is an encouraging initial finding. It suggests that accurate BHIP prediction may not require sensitive criminal justice data. While this is promising evidence, one must acknowledge that pattern does not, by itself, guarantee fairness. The text-derived behavioral indicators that do drive the model could themselves reflect implicit biases in how encounters are documented for different populations. Therefore, we proceed with a more formal fairness analysis below, where we directly evaluate the model's error rates and predictive accuracy across racial groups.

Leave-Out Experiments

To experimentally validate the insights from the SHAP analysis—namely, that predictive power is driven by ambulance data rather than police or demographic features—we conducted a series of feature exclusion experiments. We iteratively rebuilt the ML models, systematically excluding specific families of features, and then evaluated the impact on predictive performance. The models explored were: (1) the full model, (2) a model that excludes arrest data (labeled "no_pd"), (3) a model that excludes both arrest and demographic data (labeled "no_pd_no_demog"), and (4) a FasterRisk (Liu et al. 2022) model that applies systematic interpretable modeling methodology to 20 of the most influential features identified through SHAP analysis.[7]

The FasterRisk approach optimizes sparse linear models with integer coefficients. We constrained the solution to select a maximum of 6 features to promote memorability and practical implementation. Table 13 presents the two simple scoring models identified by this process. The feature selection was largely consistent across both folds—resulting in a three-feature model for Fold 1 and a four-feature model for Fold 2—with both models selecting a core set of indicators derived from paramedic text fields: the count of prior BHIPs, events with police presence, and events with psychiatric indicators.

Table 4 and Fig. 2 present the PPV and Sensitivity for these ablated LightGBM models and the simple FasterRisk approach across various identification thresholds. Results for

[7] We used the publicly available Python implementation of the FasterRisk algorithm.

Table 4 Grouped ablation performance for LightGBM

Model	Top 0.1%	Top 1%	Top 2.5%	Top 5%
PPV				
Full Model	0.508	0.202	0.117	0.075
	(0.437-0.583)	(0.183-0.220)	(0.108-0.126)	(0.069-0.080)
No Police Data	0.497	0.200	0.118	0.075
	(0.432-0.578)	(0.182-0.219)	(0.108-0.127)	(0.069-0.080)
No Police or Demographic Data	0.503	0.202	0.117	0.075
	(0.432-0.573)	(0.184-0.221)	(0.108-0.126)	(0.070-0.080)
FasterRisk	0.487	0.197	0.107	0.069
	(0.407-0.568)	(0.179-0.215)	(0.098-0.116)	(0.064-0.074)
Sensitivity				
Full Model	0.067	0.267	0.387	0.493
	(0.058-0.076)	(0.246-0.287)	(0.363-0.410)	(0.469-0.517)
No Police Data	0.065	0.265	0.389	0.493
	(0.057-0.076)	(0.244-0.285)	(0.365-0.413)	(0.467-0.519)
No Police or Demographic Data	0.066	0.267	0.386	0.497
	(0.057-0.076)	(0.245-0.288)	(0.361-0.409)	(0.471-0.519)
FasterRisk	0.064	0.261	0.354	0.458
	(0.054-0.074)	(0.240-0.280)	(0.331-0.376)	(0.433-0.482)

Note: Table shows the performance of LightGBM and FasterRisk models, reporting both PPV and Sensitivity at different identification thresholds (Top 0.1%, 1%, 2.5%, and 5%). The LightGBM models include a Full Model and ablated versions with No Police Data and No Police or Demographic Data. FasterRisk results represent performance calculated on pooled out-of-sample predictions from systematic interpretable models trained separately on Fold 1 and Fold 2, each with access to the 20 most influential features identified through SHAP analysis. For each model and threshold, the table provides the point estimate and its corresponding 95% bootstrapped confidence interval (1,000 replications) in parentheses

the Random Forest and Logistic Regression model are similar and reported in Appendix Tables 7 and 8, respectively. The results indicate that excluding police data or both police and demographic data has a negligible impact on the model's performance. The "No Police Data" and "No Police or Demographic Data" models maintain PPV and Sensitivity metrics that are nearly identical to those of the "Full Model" across all thresholds. This confirms the finding from our SHAP analysis that the model's predictive power is derived primarily from the rich ambulance data, and it suggests that fairness-related exclusions of police arrest or demographic information may not require a substantial trade-off in predictive accuracy.

The FasterRisk approach, despite the simplicity of the prediction models identified, performs remarkably close to the complex "black box" LightGBM models. At the 1% threshold, for example, its PPV of 0.197 is statistically indistinguishable from the Full Model's PPV of 0.202. This suggests that a streamlined, transparent approach can achieve most of the predictive gains of a more complex model, offering a practical path toward implementation in high-stakes contexts.

Demographics of Identified Populations

Having established the features that drive the model's predictions, we now turn to the characteristics of the individuals each approach identifies as high-risk. Table 5 shows demographic attributes and service encounters for the top 1% of individuals identified by each

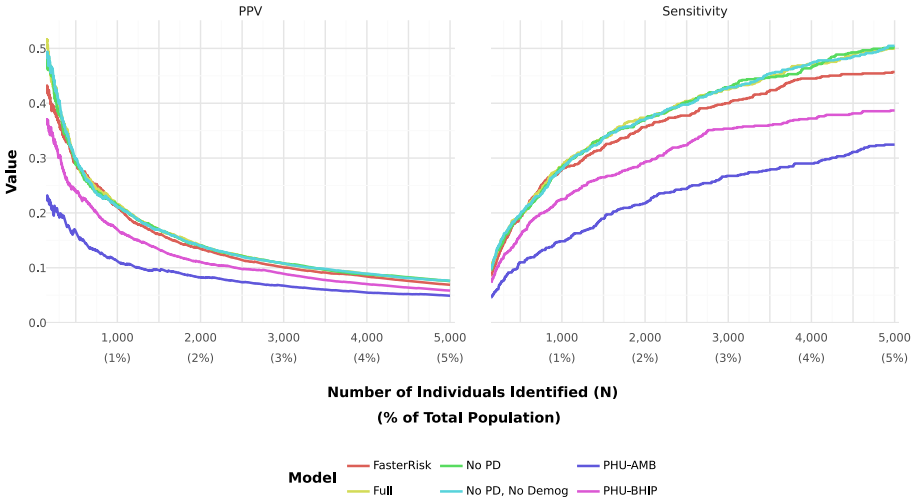


Fig. 2 Predictive performance. Left panel shows PPV and right panel shows sensitivity across a range of individuals identified. To mimic real world scenario in which only one simple model is used, metrics are reported for out-of-sample predictions for Fold 1 ($N = 99,635$). The models include: a) Full - uses all 283 features, b) No PD - drops arrest data, c) No PD, No Demog - additionally drops demographic features, d) FasterRisk - the three variable model from the simple modeling process e) PHU-AMB - uses prior ambulance run counts to rank f) PHU-BHIP - uses prior BHIP to rank. Except for high user results (PHU-BHIP and PHU-AMB) and FasterRisk, all models are LightGBM

model. The baseline approaches identify distinct populations. The PHU-BHIP strategy identifies a younger group (average age 39.4) that, by definition, consists entirely of individuals with at least one prior BHIP. The PHU-AMB model, which ranks by ambulance use, identifies an older group (average age 50.6) that is disproportionately Black (69% vs. 56% overall) and has the highest rate of indicated homelessness (32%).

The machine learning models identify populations with even higher concentrations of service needs. Individuals identified by the Full LGB-ML and FasterRisk models are more likely than those in the PHU-BHIP group to have indications of mental health issues, substance use, and aggression. However, a key trade-off for the simplicity and interpretability of the FasterRisk model appears in its capacity for identifying at-risk individuals who have no BHIPs in the last year. Tables 3 and 5 show that while 24% of individuals in the top 1% risk group identified by the Full LGB-ML model had no prior BHIP, this figure is lower for the FasterRisk model, at 15%.

Fairness Analysis Across Racial Groups

Our preceding analyses revealed that while the groups identified by the machine learning models are disproportionately Black compared to the overall sample, the models' predictions are not driven by demographic or arrest features. In this section, we assess whether the models operate equitably, as reflected through three common notions of fairness: (1) whether lower-risk individuals from different groups are flagged incorrectly at similar rates (False Positive Rate (FPR) parity); (2) whether a high-risk prediction is equally accurate

Table 5 Characteristics of the Top 1% of individuals identified by each approach

Characteristic	Overall	Full LGB-ML	FasterRisk	PHU-BHIP	PHU-AMB
Age	47.6	41.6	41.1	39.4	50.6
		(41.0-42.2)	(40.5-41.7)	(38.9-40.2)	(50.0-51.3)
Male	0.57	0.63	0.65	0.60	0.62
		(0.61-0.66)	(0.63-0.67)	(0.59-0.64)	(0.59-0.64)
Black	0.56	0.59	0.57	0.51	0.69
		(0.57-0.62)	(0.55-0.60)	(0.50-0.54)	(0.66-0.71)
Hispanic	0.16	0.14	0.14	0.17	0.10
		(0.12-0.15)	(0.13-0.16)	(0.15-0.18)	(0.08-0.11)
White	0.23	0.24	0.25	0.27	0.20
		(0.22-0.26)	(0.23-0.27)	(0.24-0.28)	(0.18-0.22)
Homeless	0.02	0.28	0.29	0.15	0.32
		(0.26-0.30)	(0.27-0.31)	(0.13-0.17)	(0.30-0.34)
Mental Health	0.19	1.00	0.97	0.89	0.71
		(1.00-1.00)	(0.97-0.98)	(0.87-0.90)	(0.69-0.73)
Substance Use	0.13	0.43	0.44	0.33	0.47
		(0.41-0.46)	(0.42-0.47)	(0.31-0.35)	(0.44-0.49)
Aggression	0.04	0.56	0.55	0.46	0.31
		(0.53-0.58)	(0.53-0.58)	(0.45-0.50)	(0.28-0.32)
BH-involved	0.13	1.00	0.99	1.00	0.56
		(1.00-1.00)	(0.98-0.99)	(1.00-1.00)	(0.54-0.59)
Police-involved	0.14	0.85	0.94	1.00	0.46
		(0.84-0.87)	(0.92-0.95)	(1.00-1.00)	(0.43-0.47)
BHIP	0.03	0.76	0.85	1.00	0.22
		(0.74-0.78)	(0.84-0.87)	(1.00-1.00)	(0.21-0.24)

Note: For each model, the top 1% of individuals at highest risk are identified. Table shows the mean value for each characteristic. The upper panel presents demographic characteristics (Age, Male, Black, Hispanic, White). The lower panel details characteristics related to prior service encounters and behavioral health indicators (Homeless, Mental Health, Substance Use, Aggression, BH-involved, Police-involved, BHIP). The first column shows prevalence rates for the full sample. For model-identified groups, the 95% confidence interval from 1,000 bootstrap replications is shown in parentheses in the row below the point estimate

regardless of group (PPV parity); and (3) whether at-risk individuals are missed by the model at similar rates across groups (False Negative Rate (FNR) parity).[8]

Table 6 presents the results of this analysis, PPV, FPR, and FNR for Black, White, and Hispanic individuals across models. For reference, the share of individuals of each race who have a BHIP in the outcome period for these groups are 0.9%, 0.6%, and 0.8%, respectively. The analysis of the LightGBM and FasterRisk models reveals several key patterns. First, FPR for the machine learning models are low and nearly identical across racial groups. At the 1% risk threshold, for example, the FPRs for Black, White, and Hispanic individuals

[8] We acknowledge that these three metrics, while common and directly relevant to the practical application of a risk model, do not represent an exhaustive list of all relevant fairness definitions. The literature on algorithmic fairness has highlighted the complexity of this issue, including the mathematical trade-offs between different fairness criteria (Chouldechova 2017) and other important concepts such as individual fairness (treating similar individuals similarly) and procedural fairness (Grgić-Hlača et al. 2018) in how a model is ultimately used. Our analysis focuses on these specific group fairness metrics because they are directly interpretable and align with the goal of understanding error rates and predictive accuracy across different communities.

Table 6 Fairness metrics by race, model, and prediction threshold

Race	Top 0.1%	Top 1%	Top 5%	Race	Top 0.1%	Top 1%	Top 5%
LightGBM				PHU-BHIP			
PPV				*PPV*			
Black	0.546	0.227	0.086	Black	0.447	0.166	0.060
White	0.419	0.153	0.053	White	0.365	0.090	0.037
Hispanic	0.500	0.205	0.078	Hispanic	0.485	0.147	0.056
FPR				*FPR*			
Black	0.001	0.008	0.044	Black	0.001	0.008	0.043
White	0.001	0.009	0.055	White	0.001	0.011	0.057
Hispanic	0	0.007	0.044	Hispanic	0.001	0.010	0.049
FNR				*FNR*			
Black	0.926	0.719	0.514	Black	0.947	0.826	0.680
White	0.933	0.729	0.487	White	0.929	0.818	0.636
Hispanic	0.950	0.762	0.512	Hispanic	0.933	0.787	0.621
N				*N*			
Black	130	1,187	5,390	Black	114	1,003	5,076
White	43	478	2,622	White	52	543	2,668
Hispanic	24	278	1,492	Hispanic	33	348	1,624
FasterRisk				PHU-AMB			
PPV				*PPV*			
Black	0.569	0.219	0.081	Black	0.252	0.105	0.047
White	0.352	0.144	0.048	White	0.205	0.112	0.044
Hispanic	0.385	0.218	0.073	Hispanic	0.368	0.136	0.062
FPR				*FPR*			
Black	0	0.008	0.044	Black	0.001	0.011	0.059
White	0.001	0.009	0.055	White	0.001	0.008	0.041
Hispanic	0.001	0.007	0.047	Hispanic	0	0.006	0.030
FNR				*FNR*			
Black	0.931	0.735	0.556	Black	0.964	0.849	0.664
White	0.929	0.732	0.532	White	0.967	0.836	0.680
Hispanic	0.958	0.742	0.521	Hispanic	0.971	0.887	0.742
N				*N*			
Black	116	1,156	5,255	Black	135	1,371	6,820
White	54	501	2,616	White	44	393	1,955
Hispanic	26	285	1,574	Hispanic	19	199	1,002

Note: Table presents disaggregated performance metrics by racial group (Black, White, Hispanic) for each predictive model (LightGBM, FasterRisk, PHU-BHIP, PHU-AMB). For each model and racial group, it displays Positive Predictive Value (PPV), False Positive Rate (FPR), False Negative Rate (FNR), and the count of individuals (N) identified within the top 0.1%, 1%, 2.5%, and 5% risk thresholds. PHU = Prior High-User

vary by less than 0.002 for both LightGBM and FasterRisk. This indicates that no single racial group is disproportionately burdened by incorrect high-risk classifications .[9]

Second, the models' predictions are often more accurate for minority groups. For both LightGBM and FasterRisk, the PPV for Black and Hispanic individuals is consistently higher than for White individuals. For LightGBM at the 1% threshold, the PPV for Black

[9] Corresponding fairness metrics for the Random Forest and Logistic Regression models are reported in Appendix Table 9.

individuals is 0.227, compared to 0.153 for White individuals. Finally, the FNR for the ML models are not only comparable across racial groups but are also substantially lower than the FNRs of the PHU-BHIP baseline, showing the ML models miss fewer at-risk individuals across all groups.

The combination of consistent FPRs across groups, alongside a higher PPV for Black and Hispanic individuals, suggests these patterns reflect underlying differences in the prevalence and nature of behavioral health crises in the data, rather than straightforward algorithmic bias that targets or mis-classifies persons based on race. Notably, the difference in PPV across groups implies that our model is not perfectly calibrated by race—a high-risk classification corresponds to different actual probabilities for different groups. This is consistent with theoretical results showing that calibration parity is generally unachievable when base rates differ across groups (Chouldechova 2017).

Statistical Power

An under-appreciated benefit of ML's superior PPV is its impact on the statistical power of randomized controlled trials (RCTs) evaluating interventions. We demonstrate here how higher PPV translates to smaller required sample sizes for program evaluation, with full mathematical derivations provided in Appendix H. Consider an experimental evaluation of a hypothetical Assertive Community Treatment (ACT) program aimed at reducing BHIP encounters among high-risk individuals. The initial study design includes 200 individuals split randomly between treatment and control conditions. If these individuals were drawn from the highest-ranked individuals identified by the full LightGBM model, the outcome rate of BHIP would be 0.508 among controls (Table 2). With this sample size, one has 80% power to detect a 38% reduction in BHIP (an average of 19 fewer individuals with a BHIP in the treatment group (31/100) as compared with the control group). Using the PHU list, the outcome rate for the control group would be 0.42; one would require 310 individuals (55% larger sample) to detect the same 38% decrease.

As noted, ML's PPV and statistical power advantages increase with identified population size. With a sample size of 2,000 selected using the LightGBM list and equally split between treatment and control groups, one has statistical power to detect a 23% reduction in BHIP—an average of 46 fewer treatment-group participants with BHIP (154/1000), compared with controls (202/1000). Using the PHU model list for the same evaluation, one would require an 81% increase in sample size, 3,620 individuals, to detect the same 23% percent decrease. This hypothetical exercise demonstrates that using ML-guided risk prediction could allow us to evaluate the true effectiveness of ACT-like programs with fewer resources (e.g. smaller sample size and/or shorter follow-up durations), or to identify smaller treatment effects with greater confidence.

Discussion

This study finds that the predictive advantage of machine learning for identifying BHIP risk stems not from algorithmic sophistication, but from systematic integration of rich, multi-source administrative data—particularly the conversion of qualitative paramedic narratives

into structured predictive features. This finding has important implications for implementation: jurisdictions need not invest heavily in cutting-edge algorithms but may benefit from developing robust data integration and text processing capabilities that leverage the expertise of front-line staff.

Equally important is our demonstration that effective machine learning for high-stakes prediction requires a comprehensive methodology that extends beyond simply applying algorithms to data. We employed a systematic three-step approach: (1) training standard ML models to establish a benchmark for predictive accuracy, (2) conducting model explanation through SHAP analysis to understand which features drive predictions, and (3) systematically searching for simple, interpretable models that retain predictive power while ensuring transparency suitable for justice settings (Rudin 2019).

A critical fourth step involves human review and validation of algorithmically identified features. While SHAP analysis and FasterRisk optimization can reveal which variables drive predictions, determining whether these features are appropriate, ethical, and legitimate for practice requires domain expertise, stakeholder input, and careful consideration of potential consequences that cannot be assessed through algorithmic methods alone (Council on Criminal Justice 2024; Stapleton et al. 2022).

Beyond these methodological contributions, our findings suggest concrete pathways for prevention practice. The ability to identify individuals at risk of their first BHIP offers potential for more proactive intervention approaches. Traditional high-user approaches, while effective for identifying individuals with established patterns, exclude a substantial portion of future BHIP cases—those without recent history. Our finding that paramedic narratives contain early warning signals for escalating behavioral health crises suggests that emergency medical services may represent an underexplored resource for prevention efforts. EMS agencies might serve as partners in community behavioral health efforts, possibly helping to identify individuals for voluntary service connection before law enforcement involvement occurs, though the feasibility and effectiveness of such approaches would require careful evaluation.

The trade-offs between model complexity and implementation feasibility present agencies with a spectrum of options tailored to their technical capacity and prevention priorities. The FasterRisk approach offers an immediate "middle path"—memorizable scoring rules that require no computational infrastructure while substantially outperforming current practice. However, agencies prioritizing identification of first-time BHIP cases may find the investment in full ML deployment justified, given the 28% vs. 15% advantage in detecting individuals without prior events. This suggests that prevention-focused initiatives may require more sophisticated approaches than those aimed primarily at managing known high utilizers.

Our findings also have implications for strengthening intervention research in this domain. The statistical power gains we document—reducing required sample sizes by 33-55%—could improve the feasibility of rigorous program evaluation. These gains become particularly valuable given a fundamental challenge in this field: we currently lack proven interventions for preventing BHIP events, let alone understanding of who might benefit most from treatment (Na et al. 2015). This raises an important distinction between our approach and ideal targeting strategies. In fields with established interventions, optimal targeting would identify individuals with the highest expected treatment benefit—

those whose outcomes would improve most with intervention. However, determining heterogeneous treatment effects requires substantial prior evidence about what works and for whom. In the absence of such evidence, baseline risk serves as a pragmatic proxy for treatment prioritization, operating under the assumption that preventing high-probability events yields greater absolute benefit than preventing low-probability ones. Our risk-based targeting thus serves a foundational role in building the very evidence base that would enable more sophisticated approaches.

However, realizing these statistical advantages depends critically on successful recruitment and retention of identified high-risk individuals. Cold outreach to community members for voluntary prevention programs may face substantial recruitment challenges, as individuals may be reluctant to engage with services, particularly if they perceive stigma or lack trust in institutions (Bhatt et al. 2024; Ludwig 2004).

Conversely, contexts where individuals are already engaged with services may see higher participation rates. Offering enhanced case management to high-risk individuals already receiving mental health services, or implementing co-responder programs within existing emergency response frameworks, may achieve higher recruitment rates. Similarly, interventions embedded within justice system touchpoints—such as enhanced diversion programs for individuals with pending charges—may benefit from higher participation due to existing institutional relationships. The magnitude of any statistical advantages will therefore vary substantially based on program design, community trust, and the voluntary versus mandated nature of participation.[10]

Several additional limitations constrain the generalizability of our findings.

The single-city design limits generalizability. Police practices, EMS documentation protocols, ambulance dispatch procedures, and community demographics vary substantially across jurisdictions. Additionally, our text-mining approach depends on specific documentation patterns by paramedics in this system, which may not transfer to other EMS contexts with different narrative styles or terminology.

Our outcome definition, while comprehensive within the EMS data, excludes police-behavioral health encounters that do not involve ambulance response. This exclusion occurs because our arrest data does not contain information which would allow us to measure BHIP when EMS is not present. This limitation may be consequential in two ways. First, behavioral health encounters involving police but not ambulances may represent situations where individuals are directly transported by police to psychiatric facilities or jails, or encounters that escalate rapidly without medical intervention being called. Second, this data limitation may partially explain why police-derived features show low importance in our models—if the most police-intensive behavioral health encounters are systematically excluded from our outcome measure, then police contact history would naturally appear less predictive than it might be for a more comprehensive measure of police-behavioral health interactions.

This constraint also underscores that our findings are most directly applicable from the perspective of emergency medical services plan subset of police-behavioral health interac-

[10] Our person-level modeling approach is designed for periodic population screening—generating ranked lists for proactive outreach. An alternative encounter-triggered approach, common in pretrial risk assessment, would assess risk when individuals present to services. We pursued the former due to data constraints, though we would not expect material differences in predictive accuracy; the choice is primarily operational.

tions. For EMS administrators, community paramedics, and agencies designingning and intervention. Our models are optimized to predict which individuals known to ambulance services are at highest risk of future encounters requiring both behavioral health support and police presence—a specific but important ambulance-based prevention programs, this represents the relevant universe of cases they can systematically identify and potentially intervene with using existing data infrastructure.

The 12-month identification period for feature construction is another methodological constraint. Our "no prior BHIP" classification is specific to this window; some individuals may have had events pre-dating this period. This is particularly salient given that approximately two-thirds of individuals with a BHIP in the outcome period had no recorded BHIP in the preceding 12 months—a longer lookback period might reveal prior events for some of these individuals. Additionally, some individuals who experience BHIPs are not present in our training data at all, limiting our ability to generate predictions for them. For jurisdictions implementing similar approaches, we would recommend using a longer lookback period where data infrastructure permits.

Most critically, our reliance on keyword searches to create features from paramedic narratives introduces measurement error and potential bias. This method creates gaps between the concepts we aim to measure—such as behavioral health crises—and what our features actually capture through specific word presence. For example, narratives describing "combative" patients may indicate crises but be missed by our search terms, while "depressed" could refer to skull fractures rather than mental states.

Large Language Models (LLMs) represent a promising direction for improving upon our keyword-based approach. LLMs could capture nuanced language and context that simple pattern matching misses—for instance, distinguishing "depressed" as a mental state versus a skull fracture description. However, more sophisticated extraction methods cannot address bias in what gets documented in the first place; if documentation practices vary systematically across populations, better extraction will faithfully capture those differences. Future work should explore LLM-based feature extraction while also attending to standardizing documentation practices.

Given the importance of text-derived indicators in our models, jurisdictions might also consider developing structured protocols for paramedics to document behavioral health observations—for instance, standardized fields or prompts for noting signs of mental health crisis, substance use, or housing instability. Such protocols would need to be developed collaboratively with EMTs and their trainers to ensure they integrate smoothly into existing workflows without compromising patient care.

This work illuminates the trade-offs facing jurisdictions considering enhanced risk prediction: between accuracy and simplicity, between predictive power and transparency, and between leveraging available data and avoiding potential bias. While our findings suggest meaningful improvements over current practice are achievable, the ultimate value of any predictive approach will depend not just on statistical performance, but on whether it enables more effective, equitable, and acceptable interventions for individuals experiencing behavioral health crises.

A Appendix - Tables

Table 7 Grouped ablation performance for random forest

Model	Top 0.1%	Top 1%	Top 2.5%	Top 5%
PPV				
Full Model	0.492	0.196	0.117	0.075
	(0.422-0.558)	(0.179-0.214)	(0.108-0.127)	(0.070-0.080)
No Police Data	0.467	0.193	0.115	0.073
	(0.387-0.533)	(0.176-0.212)	(0.107-0.125)	(0.068-0.078)
No Police or Demographic Data	0.472	0.193	0.115	0.073
	(0.387-0.533)	(0.175-0.211)	(0.106-0.125)	(0.068-0.079)
FasterRisk	0.487	0.197	0.107	0.069
	(0.407-0.568)	(0.179-0.215)	(0.098-0.116)	(0.064-0.074)
Sensitivity				
Full Model	0.065	0.259	0.388	0.493
	(0.056-0.073)	(0.238-0.279)	(0.364-0.411)	(0.469-0.518)
No Police Data	0.062	0.255	0.381	0.481
	(0.051-0.070)	(0.234-0.276)	(0.358-0.407)	(0.459-0.508)
No Police or Demographic Data	0.062	0.255	0.380	0.481
	(0.050-0.070)	(0.234-0.276)	(0.355-0.404)	(0.457-0.507)
FasterRisk	0.064	0.261	0.354	0.458
	(0.054-0.074)	(0.240-0.280)	(0.331-0.376)	(0.433-0.482)

Note: Table shows the performance of Random Forest and FasterRisk models, reporting both PPV and Sensitivity at different identification thresholds (Top 0.1%, 1%, 2.5%, and 5%). The Random Forest models include a Full Model and ablated versions with No Police Data and No Police or Demographic Data. FasterRisk results represent performance calculated on pooled out-of-sample predictions from systematic interpretable models trained separately on Fold 1 and Fold 2, each with access to the 20 most influential features identified through SHAP analysis. For each model and threshold, the table provides the point estimate and its corresponding 95% bootstrapped confidence interval (1,000 replications) in parentheses

Table 8 Grouped ablation performance for logistic regression

Model	Top 0.1%	Top 1%	Top 2.5%	Top 5%
PPV				
Full Model	0.392	0.190	0.114	0.076
	(0.337-0.472)	(0.173-0.207)	(0.105-0.123)	(0.070-0.081)
No Police Data	0.417	0.190	0.113	0.074
	(0.357-0.488)	(0.172-0.207)	(0.104-0.121)	(0.069-0.079)
No Police or Demographic Data	0.417	0.190	0.113	0.074
	(0.357-0.488)	(0.172-0.207)	(0.104-0.121)	(0.069-0.079)
FasterRisk	0.487	0.197	0.107	0.069
	(0.407-0.568)	(0.179-0.215)	(0.098-0.116)	(0.064-0.074)
Sensitivity				
Full Model	0.052	0.251	0.376	0.499
	(0.044-0.062)	(0.231-0.271)	(0.354-0.401)	(0.475-0.525)
No Police Data	0.055	0.251	0.372	0.489
	(0.047-0.064)	(0.231-0.270)	(0.348-0.395)	(0.463-0.516)
No Police or Demographic Data	0.055	0.251	0.372	0.489
	(0.047-0.064)	(0.231-0.270)	(0.348-0.395)	(0.463-0.516)
FasterRisk	0.064	0.261	0.354	0.458
	(0.054-0.074)	(0.240-0.280)	(0.331-0.376)	(0.433-0.482)

Note: Table shows the performance of Logistic Regression and FasterRisk models, reporting both PPV and Sensitivity at different identification thresholds (Top 0.1%, 1%, 2.5%, and 5%). The Logistic Regression models include a Full Model and ablated versions with No Police Data and No Police or Demographic Data. FasterRisk results represent performance calculated on pooled out-of-sample predictions from systematic interpretable models trained separately on Fold 1 and Fold 2, each with access to the 20 most influential features identified through SHAP analysis. For each model and threshold, the table provides the point estimate and its corresponding 95% bootstrapped confidence interval (1,000 replications) in parentheses

Table 9 Fairness metrics by race, model, and prediction threshold

Race	Top 0.1%	Top 1%	Top 2.5%	Top 5%
Random Forest				
Precision (PPV)				
Black	0.565	0.221	0.132	0.085
White	0.353	0.143	0.086	0.055
Hispanic	0.455	0.202	0.128	0.079
FPR				
Black	0	0.008	0.022	0.045
White	0.001	0.009	0.025	0.053
Hispanic	0	0.007	0.021	0.044
FNR				
Black	0.927	0.728	0.612	0.515
White	0.933	0.736	0.606	0.476
Hispanic	0.958	0.767	0.608	0.504
N				
Black	124	1,176	2,809	5,431
White	51	495	1,239	2,570
Hispanic	22	277	736	1508
Logistic Regression				
Precision (PPV)				
Black	0.441	0.210	0.122	0.079
White	0.227	0.140	0.089	0.061
Hispanic	0.462	0.199	0.133	0.088
FPR				
Black	0.001	0.009	0.023	0.049
White	0.001	0.009	0.024	0.048
Hispanic	0	0.007	0.019	0.041
FNR				
Black	0.941	0.731	0.623	0.514
White	0.963	0.762	0.613	0.472
Hispanic	0.950	0.779	0.617	0.487
N				
Black	127	1,223	2,958	5,896
White	44	456	1,174	2,319
Hispanic	26	266	692	1,400

Note: For each model, the table shows Precision (PPV), False Positive Rate (FPR), False Negative Rate (FNR), and the count of individuals (N) from each racial group within the top k% at the specified threshold. PHU = Prior High-User

Table 10 Detailed breakdown of predictive features by category

Feature Category	Sub-Category / Description	Count
Demographics		**7**
	Gender (one-hot encoded)	3
	Race (one-hot encoded)	4
PD-Derived	*(Features from Police Department Arrest Data)*	**68**
	Frequency (Counts)	
	Total Arrests	1
	Counts by Charge Class (Full Period)	4
	Counts by FBI Code (Full Period)	28
	Counts by Charge Class (Last 90 Days)	4
	Counts by FBI Code (Last 90 Days)	28
	Recency (Timing)	
	Time Since First & Last Arrest	2
	Persistence	
	Unique Months with an Arrest	1
FD-Derived (Non-Text)	*(Features from structured ambulance data)*	**4**
	Frequency (Total Ambulance Runs)	1
	Recency (Time Since First & Last Run)	2
	Persistence (Unique Months with a Run)	1
FD-Derived (Text)	*(Features from 56 text-mined flags)*	**168**
	Frequency (1 count per flag)	56
	Recency (Time since most recent appearance of flag)	56
	Recency (Time since first appearance of flag)	56
Combined Persistence	Unique Months with Any Contact (PD or FD)	**1**
Modal Temporal Patterns	Most Frequent Season and Time of Day of Encounters	**35**
Total Features (Counted)		**283***

Note: Bold entries indicate feature category headers and their corresponding count subtotals, distinguishing them from individual sub-category rows beneath each header

Table 11 High risk-group size by model for subgroups with and without prior BHIPs

Model	Top 0.1%	Top 1%	Top 2.5%	Top 5%
With Prior BHIP				
N in Top Group				
LightGBM	191	1,518	3,022	4,085
	(185-196)	(1,480-1,559)	(2,949-3,101)	(3,978-4,191)
FasterRisk	193	1,705	4,467	5,583
	(188-198)	(1,674-1,735)	(4,419-4,506)	(5,441-5,736)
PHU-BHIP	199	1,996	4,991	5,583
	(199-199)	(1,996-1,996)	(4,991-4,991)	(5,433-5,733)
Without Prior BHIP				
N in Top Group				
LightGBM	8	478	1,969	5,898
	(3-14)	(437-516)	(1,889-2,042)	(5,791-6,005)
FasterRisk	6	291	524	4,400
	(1-11)	(260-322)	(485-571)	(4,246-4,541)
PHU-BHIP	0	0	0	4,400
	(0-0)	(0-0)	(0-0)	(4,249-4,549)

Note: For each subgroup, the table shows the sample size from the top-ranked group across various identification thresholds (Top 0.1%, 1%, 2.5%, and 5%). Point estimates are presented, with 95% confidence intervals from 1,000 bootstrap replications shown in parentheses. The PHU-BHIP model by definition does not identify individuals without prior BHIPs at certain thresholds

B Appendix - Figures

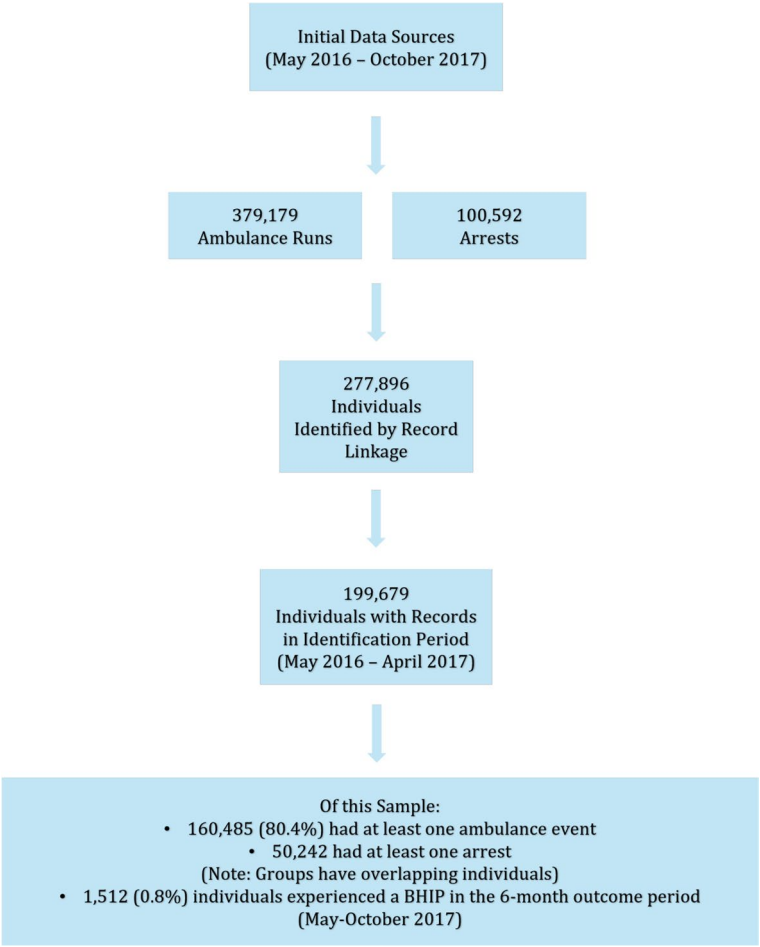


Fig. 3 Figure shows sample construction steps

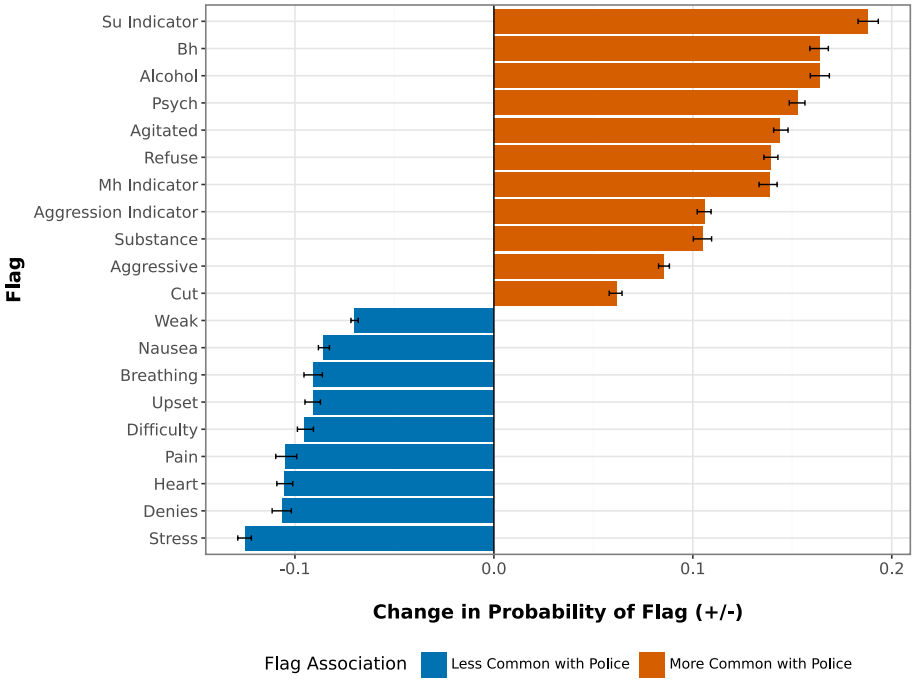


Fig. 4 Figure shows the increase (or decrease) in the probability of an incident exhibiting the flag type shown on the y-axis between incidents that are flagged and not flagged as having police presence. Only flags are shown where the magnitude of this difference is greater than five percentage points. Bootstrapped 95% confidence intervals are shown in black

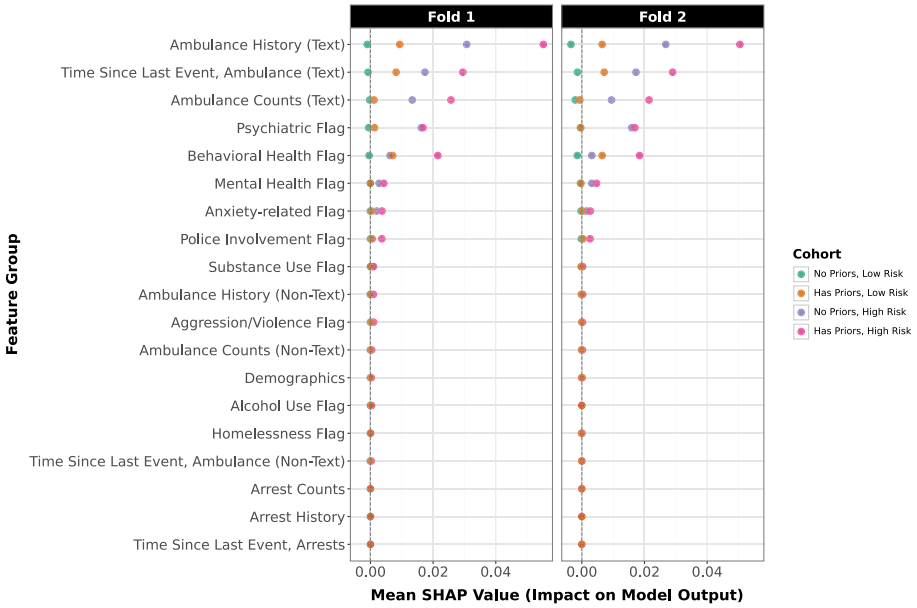


Fig. 5 Mean SHAP (SHapley Additive exPlanations) values for random forest model features, stratified by risk and prior BHIP status. This plot illustrates the average impact (in log odds) of different feature groups on the Random Forest model's output. Features are grouped by their type and ranked by their overall mean SHAP value. The plot is split into two folds (Fold 1 and Fold 2) to show consistency of feature importance across different data partitions. Within each fold, points are colored to represent four distinct cohorts: 'No Priors, Low Risk' (green), 'Has Priors, Low Risk' (teal), 'No Priors, High Risk' (purple), and 'Has Priors, High Risk' (orange). 'Priors' refers to whether an individual had a prior BHIP encounter during the identification period

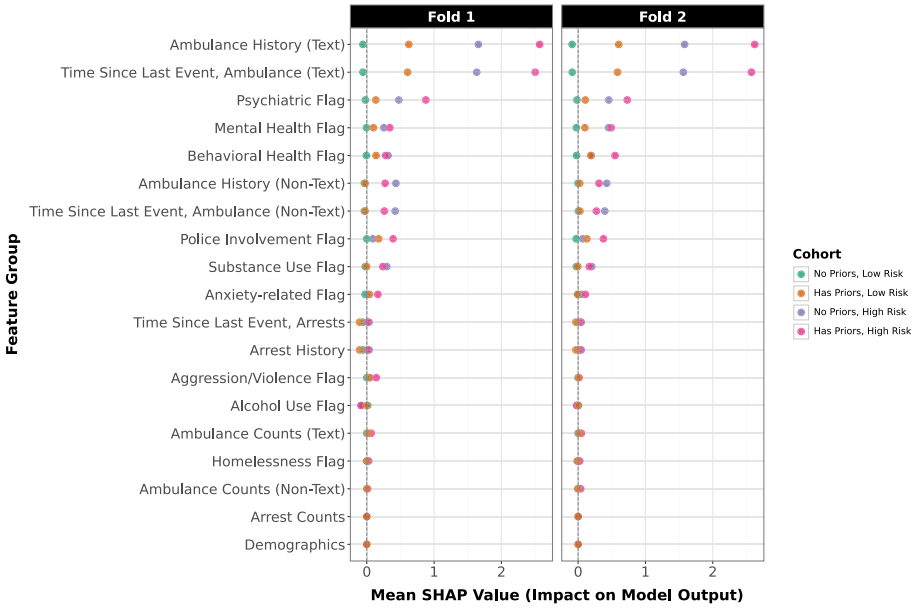


Fig. 6 Mean SHAP (SHapley Additive exPlanations) values for logistic regression model features, stratified by risk and prior BHIP status. This plot illustrates the average impact (in log odds) of different feature groups on the Logistic Regression model's output. Features are grouped by their type and ranked by their overall mean SHAP value. The plot is split into two folds (Fold 1 and Fold 2) to show consistency of feature importance across different data partitions. Within each fold, points are colored to represent four distinct cohorts: 'No Priors, Low Risk' (green), 'Has Priors, Low Risk' (teal), 'No Priors, High Risk' (purple), and 'Has Priors, High Risk' (orange). 'Priors' refers to whether an individual had a prior BHIP encounter during the identification period

C Appendix - Linking

Given the realities of first-response, data recording is often error prone. Demographic data recorded for the same person across multiple events may differ because of transcription or other errors (e.g., "Jonathon Kaufman" vs. "Jonathon Koffman" in separate events). Thus, the common practice of linking records by exact matching on name and birth date can lead to a large rate of missed matches,12 and corresponding undercounts of the number of events attributable to each person.

Linking error may be correlated with both demographic attributes and with BHIP risk. For example, if individuals experiencing homelessness are more likely to experience BH-involved police encounters and have their attributes recorded differently across events, a prior BHIP approach based on counts may rank these individuals lower than other individuals who face similar risks yet have more accurate data. We therefore aim to reduce matching error via probabilistic linking considering name, date of birth, sex, race, and address for all events in our data.

Importantly, each police arrest record includes an Identification Record (IR) number, a fingerprint-verified unique individual identifier. IR numbers provide our 'ground truth'. When two arrest records have the same IR number, we know the same individual is the subject of both records, even given variations in recorded demographic data. We use co-occurrence of the IR number, and variation in recording of demographic data across arrest records for the same person to build a supervised record linkage model.[11]

[11] Linkage software available at [masked].

We can apply this model to compare any two records (two arrest records, two ambulance records, or one ambulance and one arrest record) to find a probability that the two events involve the same person, and then retain only the highest probability matches as belonging to the same individual. This technique provides a count of unique individuals across police and fire event data. We then attribute each ambulance and arrest event to one of these individuals.

D Appendix - Measuring the Outcome

Although arrest data are seemingly appropriate for measuring our BHIP outcomes, these may not to be the best data source for two reasons. First, there is no reliable indicator of behavioral or mental health status in PD arrest data. Second, most police responses do not end in arrest.

To overcome these limitations, we rely on text fields compiled by FD ambulance staff at each ambulance run. These fields capture information on the patient's complaint, symptoms, medical history, impressions of the medic, and narrative description of the event as a whole. We worked closely with FD staff to develop explicit logic using these fields to flag ambulance events as BH-involved, and to flag ambulance events where police were present.

- *Measuring BH-involved encounters*: In partnership with FD operational and data experts, we implemented keyword searches on the "complaint" and "impression" fields to identify ambulance runs that involved a community member experiencing potential BH crisis. These encounters will hereafter be referred to as BH-involved encounters. We labeled FD incidents as BH-involved if either of the following two conditions were met:The "complaint" text field contained any of the following words: agitated, anxiety, crying, depressed, distressed, fear, hallucination, hostile, irritable, nervous, panic, stress, suicidal, tearful, violent
- The "impression" text field contained any of the following words: agitated, anxiety, behavioral, combative, depression, psychiatric, suicidal, taser injury*Measuring police presence*: FD narrative fields provide valuable, albeit incomplete, information on when police are also on scene. As shown in SM.2 below, we search patient symptom, patient complaint, medic impression, event type and event narrative fields for presence of the following keywords: "[masked] PD", "police", "[masked] P.D.", or "[masked] p d".

Of the 40,593 ambulance runs (11% of all runs) with above-indications of police presence, in only 2,169 (5.3%) was there an associated arrest. Using arrests as a measure of police presence would thus vastly understate the proportion of ambulance runs that include officers on scene. (It was even more rare to observe an associated arrest when there was no indicated police presence in FD data. Of the 338,586 ambulance runs in which there was no indicated police presence, only 784 (0.2%) included an associated arrest.)

E Appendix - Prior High Use

This table shows the minimum and average number of BH-involved encounters with police for individuals ranked in the top 0.1%, 1%, and 5% of the High-user approach. Only 2.8% of individuals have one or more BH-involved encounters with police, so

Table 12 BH-involved encounters with police*

Higher-User Ranking	Minimum pBHIP	Average pBHIP
In Top 0.1%	2	3.94
In Top 1%	1	1.51
In Top 5%	0	0.66

to identify the top 5% of individuals based on this criteria, we would need to select everyone with 1+ encounter, and choose the remaining 2.2

Note that the minimum number of events for inclusion in the top 5% is zero. This points to another limitation of the High-user approach. If the event of interest is rare–as it is here–the number of individuals with nonzero prior "uses" that can be identified as high risk will be correspondingly low, thus limiting the potential impact and scope of any associated program.

F Appendix - Stability of Metrics from Pooled Predictions

F.1 Motivation

In behavioral health and criminal justice applications, interventions operate under fixed capacity constraints. For instance, a program with 50 slots must identify exactly 50 high-risk individuals from a population of 2,000. When using nested cross-validation to evaluate predictive models, a key methodological choice arises: how to compute performance metrics like positive predictive value (PPV) from the outer folds.

Two approaches exist:

- **Approach 1 (Fold-wise averaging)**: Compute PPV separately within each fold, then average
- **Approach 2 (Pooling)**: Pool all out-of-sample predictions, then compute PPV onceWe adopt the pooling approach as it directly mirrors operational deployment—selecting the top n individuals from the full population. This appendix provides the statistical rationale for this choice.

F.2 Statistical Framework

Consider nested cross-validation with:

- N = total population size
- K = number of outer folds
- n = intervention capacity (number of slots)
- p = true event rate among selected high-risk individualsBoth approaches use identical out-of-sample predictions from the outer loop. The difference lies solely in metric computation.

To illustrate the variance properties, we use the fact that PPV is a sample proportion of binary outcomes. For a sample of size *m* with true event rate *p*, the variance of the sample proportion follows the binomial distribution:

$$\mathrm{Var}(\text{proportion}) = \frac{p(1-p)}{m}$$

We use this relationship to analyze both approaches.

Approach 1: Fold-Wise Averaging

For each outer fold *k*:

1. Select the top *n*/*K* individuals within that fold
2. Compute PPV on these *n*/*K* individuals: $\hat{M}_k$
3. Average across folds: $\bar{M} = \frac{1}{K}\sum_{k=1}^{K}\hat{M}_k$Under the binomial framework, each fold-specific PPV has variance:

$$\mathrm{Var}(\hat{M}_k) = \frac{p(1-p)}{n/K} \equiv \sigma^2_{\text{fold}}$$

The variance of the average is:

$$\mathrm{Var}(\bar{M}) = \frac{1}{K^2}\left[\sum_{k=1}^{K}\mathrm{Var}(\hat{M}_k) + \sum_{i\neq j}\mathrm{Cov}(\hat{M}_i, \hat{M}_j)\right]$$

With positive correlation between fold metrics ($\mathrm{Cov}(\hat{M}_i, \hat{M}_j) \approx c_M > 0$) due to overlapping training data:

$$\mathrm{Var}(\bar{M}) \approx \frac{\sigma^2_{\text{fold}}}{K} + \frac{K-1}{K}c_M$$

The positive covariance c_M arises because models for different folds train on overlapping data (sharing $(K-2)/(K-1)$ of their training sets).

Approach 2: Pooling

Steps:

1. Pool all *N* out-of-sample predictions
2. Select the global top *n* individuals
3. Compute PPV once: M_{pool}Under the same binomial framework, M_{pool} is computed on *n* individuals:

$$\mathrm{Var}(M_{\text{pool}}) = \frac{p(1-p)}{n}$$

Expressing this in terms of σ^2_{fold}:

$$\text{Var}(M_{\text{pool}}) = \frac{p(1-p)}{n} = \frac{1}{K} \cdot \frac{p(1-p)}{n/K} = \frac{\sigma^2_{\text{fold}}}{K}$$

F.3 Comparison and Implications

Comparing variances:

- **Pooling**: $\text{Var}(M_{\text{pool}}) \approx \frac{\sigma^2_{\text{fold}}}{K}$
- **Averaging**: $\text{Var}(\bar{M}) \approx \frac{\sigma^2_{\text{fold}}}{K} + \frac{K-1}{K} c_M$The pooling approach yields lower variance when $c_M > 0$, which is typical due to training set overlap. The variance reduction is most pronounced for larger *K* values (e.g., $K = 5$ or $K = 10$).

For our implementation with $K = 2$ outer folds, the training sets are disjoint (no overlap), minimizing c_M. However, we maintain the pooling approach since inter-fold correlation may still exist due to, for example, consistency in relationships across the fold.

G Appendix - FasterRisk Feature Set and Model Specification

The FasterRisk model was trained on 20 features selected based on SHAP analysis results, representing the most influential predictors from the full 283-feature set. These features are organized into two primary categories: counts and time since most recent event for specific behavioral indicators: :

- Ambulance events
- Alcohol use
- Behavioral health indicators
- Behavioral health indicators with police presence
- Mental health indicators
- Police presence
- Psychiatric indicators
- Stress/crisis indicators
- Substance use indicators
- General substance useAll features derive exclusively from Fire Department ambulance data, with behavioral indicators extracted through keyword searches of paramedic narrative text fields. No arrest or demographic features were included in this candidate set, reflecting the SHAP analysis finding that predictive power stems primarily from clinical observations documented by emergency medical personnel. We present results from the final model chosen below:

Fold 1 Risk Score Formula: $RiskScore = 4 \cdot (BHIP) + 2 \cdot (Police) + 2 \cdot (Psych)$
Fold 2 Risk Score Formula: $RiskScore = 1 \cdot (BH) + 2 \cdot (BHIP) + 2 \cdot (Police) + 1 \cdot (Psych)$

Table 13 FasterRisk model specifications for Fold 1 and Fold 2

Feature	Coefficient (Fold 1)	Coefficient (Fold 2)
Behavioral Health (BH)	0	1
Behavioral Health + Police (BHIP)	4	2
Police Presence	2	2
"Psych" Presence	2	1

H Appendix: Statistical Power Calculations for Program Evaluation

This appendix provides the mathematical details underlying the statistical power analysis presented in Section 4.7 of the main text. We demonstrate how machine learning models' superior positive predictive value (PPV) translates into enhanced statistical power for experimental evaluations of interventions targeting individuals at risk of behavioral health-involved police encounters (BHIP).

H.1 Power Calculation Framework

For a two-sample test comparing treatment and control groups with equal allocation, the required sample size per group to achieve power $(1 - \beta)$ for detecting a relative risk reduction of δ at significance level α is given by:

$$n = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 \cdot [p_0(1-p_0) + p_1(1-p_1)]}{(p_0 - p_1)^2}$$

where:

- p_0 = baseline event rate (control group)
- p_1 = treatment group event rate = $p_0(1-\delta)$
- $z_{1-\alpha/2}$ = critical value for two-tailed test at significance level α
- $z_{1-\beta}$ = critical value corresponding to power $(1-\beta)$In our context, p_0 corresponds to the PPV of the risk identification model, representing the expected BHIP rate among individuals identified as high-risk who receive no intervention.

H.2 Comparative Power Analysis: Top 0.1% Risk Group

Using the PPV estimates from Table 2, we compare sample size requirements for detecting a 38% relative risk reduction ($\delta = 0.38$) with 80% power ($\beta = 0.20$) and $\alpha = 0.05$.

LightGBM Model (PPV = 0.508):

$$\begin{aligned} p_0 &= 0.508 \\ p_1 &= 0.508 \times (1 - 0.38) = 0.315 \\ n &= \frac{(1.96 + 0.84)^2 \cdot [0.508(0.492) + 0.315(0.685)]}{(0.508 - 0.315)^2} \\ &= \frac{7.84 \cdot [0.250 + 0.216]}{0.037} = \frac{3.65}{0.037} = 99 \text{ per group} \end{aligned}$$

Total required sample: $n_{total} = 2 \times 99 = 198$ individuals

Prior High-User Model (PPV = 0.422):

$$
\begin{aligned}
p_0 &= 0.422 \\
p_1 &= 0.422 \times (1 - 0.38) = 0.262 \\
n &= \frac{(1.96 + 0.84)^2 \cdot [0.422(0.578) + 0.262(0.738)]}{(0.422 - 0.262)^2} \\
&= \frac{7.84 \cdot [0.244 + 0.193]}{0.026} = \frac{3.43}{0.026} = 132 \text{ per group}
\end{aligned}
$$

Total required sample: $n_{total} = 2 \times 132 = 264$ individuals

Sample Size Reduction: The LightGBM model requires 33% fewer participants (66 fewer individuals) to achieve the same statistical power.

H.3 Comparative Power Analysis: Top 1% Risk Group

For the broader 1% risk group, we examine detection of a 23% relative risk reduction:

LightGBM Model (PPV = 0.202):

$$
\begin{aligned}
p_0 &= 0.202 \\
p_1 &= 0.202 \times (1 - 0.23) = 0.156 \\
n &= \frac{(1.96 + 0.84)^2 \cdot [0.202(0.798) + 0.156(0.844)]}{(0.202 - 0.156)^2} \\
&= \frac{7.84 \cdot [0.161 + 0.132]}{0.002} = \frac{2.30}{0.002} = 1,000 \text{ per group}
\end{aligned}
$$

Total required sample: $n_{total} = 2,000$ individuals

Prior High-User Model (PPV = 0.129):

$$
\begin{aligned}
p_0 &= 0.129 \\
p_1 &= 0.129 \times (1 - 0.23) = 0.099 \\
n &= \frac{(1.96 + 0.84)^2 \cdot [0.129(0.871) + 0.099(0.901)]}{(0.129 - 0.099)^2} \\
&= \frac{7.84 \cdot [0.112 + 0.089]}{0.001} = \frac{1.58}{0.001} = 1,756 \text{ per group}
\end{aligned}
$$

Total required sample: $n_{total} = 3,512$ individuals

Sample Size Reduction: The LightGBM model requires 43% fewer participants (1,512 fewer individuals) compared to the PHU approach.

Acknowledgements We gratefully acknowledge the invaluable partnership and support of our city partners, whose data and insights were critical to this research. We also thank the audiences at the Association for Public Policy Analysis and Management (APPAM) and the American Public Health Association (APHA) conferences for their constructive comments on earlier versions of this work.

Funding This research was funded by the National Institute of Mental Health under grant R01MH117168. Any opinions expressed are those of the authors.

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Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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